Reading panels of vision-language models share their errors: a controlled evaluation on radiology images

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Abstract

Purpose: To measure error correlation among vision-language model (VLM) readers of radiology questions and identify its sources.

Material and methods: Four proprietary VLMs and five open-weight VLMs answered VQA-RAD, SLAKE and MedXpertQA-MM questions. We compared single readers with nine-reader independent, discussion and self-consistency panels (identical prompts, majority vote). Manipulations varied images and shared components. Perception models wrote findings reports; interpretation models answered from them. Outcomes were accuracy, error correlation and effective number of independent readers (N_{eff}), with item bootstrap 95% confidence intervals.

Results: Qwen 3.8 Flash independent and discussion panels gained 5.6 to 13.6 and 8.8 to 14.0 percentage points over the single reader, respectively. Other proprietary models' panels changed accuracy by −2.8 to 4.0 and −3.2 to 5.6, respectively. N_{eff} was 1.17 to 1.99. On SLAKE, blank images (uniform gray images of the same size) lowered single-reader accuracy by 25.2 to 40.0 percentage points. On MedXpertQA-MM, 64.2% to 76.2% of shared wrong answers (at least seven of nine readers wrong) matched blank-image majorities. Gemini 3.7 Flash reports raised accuracy by 6.1 to 16.3 percentage points relative to the other interpretation models reading the images themselves. Shared-report error correlation exceeded own-report values (0.800 to 0.990 versus 0.641 to 0.818). Roles lowered error correlation by 0.033 to 0.060 versus 0.470 to 0.492 with different models. Readers changed 41.5% to 100.0% of correct initial minority answers to wrong answers.

Conclusions: Readers built from one VLM made correlated errors. A second reading lowered error correlation most when it came from a different model.

Keywords: Multi-agent system, Vision-language model, Double reading,

Error correlation, Visual question answering, Language prior

1. Introduction

Double reading of screening mammograms increased cancer detection and recall rates compared with single reading.[1] In diagnostic radiology, a systematic review reported discrepancy rates of 0.4% to 22%, depending on the setting.[2] Radiology distinguishes perceptual and interpretive errors.[3] Here, perception is the extraction of visual findings; interpretation is the selection of an answer from the findings and medical knowledge. Pooling diagnostic judgments outperformed the best doctor when doctors had similar accuracy.[4] The value of another reader depends on both individual accuracy and error correlation. Artificial intelligence (AI) systems have also been evaluated as second readers, including a simulation of double reading in breast cancer screening.[5]

Multi-agent systems built from large language models emulate a clinical consultation. Published medical systems assigned clinical roles to several instances of a model, which answered separately before discussing their answers or voting.[6,7] A reading panel applies this arrangement to questions about radiology images. An evaluation of agent systems found that the benefit of collaboration shrank as the underlying model became more capable.[8] Another evaluation found correlated errors across language models, including models with different architectures and providers.[9] In a reading panel built from one vision-language model (VLM), every reader shares the model's weights. These include the vision encoder and the language model. Sampling and role prompts change the text of each reader's response while the model and visual evidence remain shared.

Models exploited language priors, or answer information in the question and options alone; accuracy decreased when similar images required different answers to the same question.[10] In medical visual question answering, leading multimodal models performed worse than random guessing on simple probing questions about diagnosis.[11] Analysis of medical image questions also found correct answers accompanied by flawed image comprehension.[12] Replacing an image with a blank image (a uniform gray image of the same size) lowered medical VLM accuracy by at most 6.5 percentage points.[13] A radiology data set constructed to remove such shortcuts reduced text-only accuracy to near chance.[14]

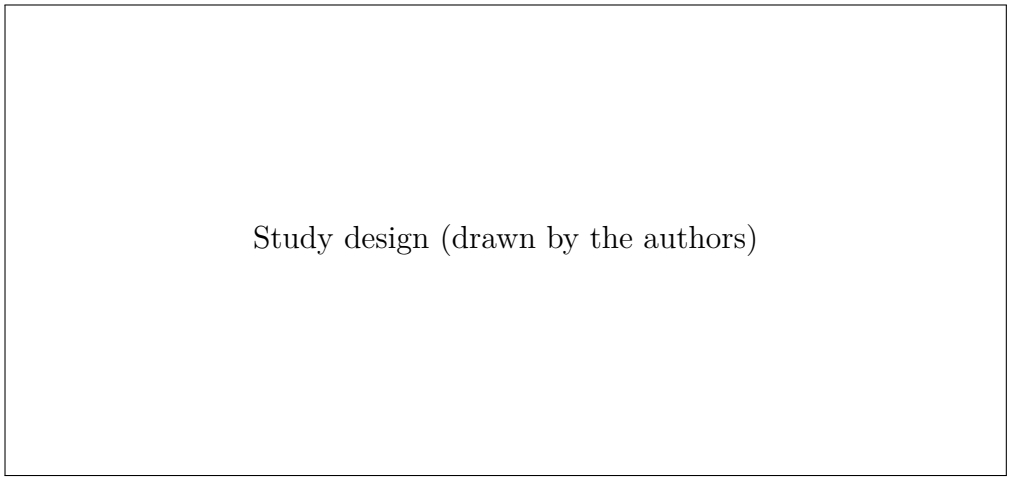


Figure 1: Study design across five evaluation sets: VQA-RAD, SLAKE, MedXpertQA-MM, SLAKE-target, and SLAKE-pairs. Comparisons varied visual evidence and components shared by single readers and reading panels. Findings reports separated perception from interpretation. Outcomes were accuracy, error correlation, and the effective number of independent readers (N_{eff}). Proprietary models: GPT-5 mini, Gemini 3.7 Flash, Claude Haiku 4.5, and Qwen 3.8 Flash. Open-weight models: MedGemma 4B, Qwen3-VL 8B, Qwen2-VL 7B, HuatuoGPT-Vision 7B, and LLaVA-1.5 7B.

The aim of this study was to measure error correlation among VLM readers answering radiology questions and identify its sources. We evaluated four proprietary VLMs and five open-weight VLMs on three data sets. These were Visual Question Answering in Radiology (VQA-RAD), the Semantically-Labeled Knowledge-Enhanced (SLAKE) data set, and multimodal MedXpertQA (MedXpertQA-MM).[15–17]

2. Material and methods

2.1. Study design

This controlled evaluation compared single readers with reading panels of VLMs. The outcomes were accuracy, error correlation, and the effective number of independent readers (N_{eff}). In the report conditions, a perception model wrote a query-agnostic findings report without the question or answer options.[18] An interpretation model answered from that report. The protocol was fixed before the main experiments.

2.2. Data sets and sampling

Evaluation sets were drawn from VQA-RAD, SLAKE, and MedXpertQA-MM (Table 1).[15–17] Duplicate VQA-RAD image–question combinations were removed before sampling. For MedXpertQA-MM, GPT-4.1 nano (gpt-4.1-nano) labeled image modality, and Claude Haiku 4.5 (claude-haiku-4.5) also labeled each candidate image as radiologic or not. Drawings, diagrams, photographs, pathology slides, charts, and tables were excluded.

SLAKE-target excluded the items of the SLAKE evaluation set. Questions concerned an organ or lesion present in the image and localized by the supplied annotations. The expanded target bounding box occupied at most 40% of the image. Each item also had an equal-sized, nonoverlapping control box with at least 70% of its area inside the body. SLAKE-pairs contained paired images with the same question and opposite reference answers. Sampling allowed at most 5 pairs per question template.

Sampling used stratified shuffling with seed 42. Strata were reference answer for VQA-RAD, reference answer and modality for SLAKE, body system and medical task for MedXpertQA-MM, and target type and modality for SLAKE-target. SLAKE-pairs was stratified by question template. Yes/no option order was randomized separately for each item using its identifier. Images were resized proportionally, without upsampling, to at most 1024 pixels on the long side.

Table 1: Evaluation sets and sampling counts.

Data set	Inclusion	Eligible	Sampled	Images	Options
VQA-RAD	Test split, closed yes/no questions	251	250	135	Yes/no
SLAKE	English test split, closed yes/no questions	355	250	93	Yes/no
MedXpertQA-MM	Every attached image radiologic; 48 items with 2–5 images	840	250	326	5
SLAKE-target	English training and test splits, closed yes/no questions on an annotated organ or lesion	335	250	196	Yes/no
SLAKE-pairs	Closed yes/no questions; 104 question templates	–	200 pairs	252	Yes/no

2.3. Models

The proprietary models were GPT-5 mini (gpt-5-mini), Gemini 3.7 Flash (gemini-3.7-flash), Claude Haiku 4.5, and Qwen 3.8 Flash (qwen3.8-flash). All reader calls were made through OpenRouter in October 2026; each condition used one sampling seed. The configuration seed was 1; reader i used seed $100 + i$. Where a model accepted these parameters, panel readers, discussion rounds, and perception models sampled at temperature 0.7 and the single reader at 0.3. GPT-5 mini, an OpenAI reasoning model, accepted neither seed nor temperature and used low reasoning effort; its readers differed through separate sampling. GPT-4.1 nano used temperature 0 for role assignment and modality screening.

Local evaluation used MedGemma 4B, Qwen3-VL 8B, Qwen2-VL 7B, HuatuoGPT-Vision 7B, and LLaVA-1.5 7B.[19–23] HuatuoGPT-Vision 7B and LLaVA-1.5 7B shared a vision encoder: the vision transformer (ViT) ViT-L/14-336 from Contrastive Language–Image Pretraining (CLIP).[24] LLaVA-Med v1.5 was excluded by the 55% pooled validation accuracy threshold (Supplementary Table S1).

2.4. Readers and reading panels

The single reader received a general-radiologist role and instructions to reason briefly, step by step, before selecting an answer. A self-consistency panel comprised 9 readers with an identical general-radiologist prompt; answers were combined by majority vote.[25] An independent panel comprised 9 readers with subspecialty roles who answered without seeing each other's answers; answers were combined by majority vote. A discussion panel comprised readers with subspecialty roles who saw each other's answers over discussion rounds. These panels followed Kim et al.'s Independent and Decentralized architectures, respectively.[8]

For each independent or discussion panel, GPT-4.1 nano selected and ranked the 9 most relevant of 12 subspecialty roles from the question text. When it returned fewer than 9 roles, the remaining positions were filled in roster order. The roster covered neuroradiology, head and neck, thoracic, cardiovascular, abdominal, genitourinary, musculoskeletal, breast, pediatric, nuclear medicine, interventional, and emergency radiology. Panels of 1, 3, and 9 readers were nested: smaller panels used the first members of the same ranked roster and reused their initial answers.

Discussion panels started from the independent panels' initial answers and allowed up to 3 subsequent rounds. Discussion stopped at unanimity.

Readers reviewed peer answers and reasoning and could retain or revise their answers. Peer context was capped at 900 characters, with peer order shuffled by item and round.

2.5. Experiment A: visual evidence and the language prior

Original images were compared with images capped at 256 or 128 pixels and with blank images. Prompts and options remained unchanged across image conditions. The single reader and independent panels were evaluated in each condition.

On SLAKE-target, the annotated target bounding box was expanded by 10% on each side and clipped to image boundaries. Target and control boxes were filled with mid-gray before resizing. On SLAKE-pairs, readers answered each member separately using the original images.

2.6. Experiment B: perception and interpretation

GPT-5 mini, Gemini 3.7 Flash, and Claude Haiku 4.5 each served as perception and interpretation models. The perception model was instructed to write at most 120 words. Reports described modality and plane alongside visible anatomy. They also recorded every abnormal or notable finding and its location. Each perception model wrote 9 reports per image or multi-image set, reused across questions about that image. All interpretation models read identical reports for a given perception model, image, and report index (1 to 9).

Interpretation models received the report, question, and options without the image. In the own-report condition, each of 9 readers received its own report; all perception–interpretation model combinations were evaluated. In the shared-report condition, all readers received the first report; this condition used matching perception and interpretation models. Each interpretation model was compared with its own self-consistency panel reading the images. For SLAKE, report coverage of the queried organ or lesion was assessed using a synonym list based on SLAKE labels.

2.7. Experiment C: shared components

For contrast adjustment, self-consistency readers 1–3 viewed original images and readers 4–6 viewed gamma-transformed images with gamma 0.7. Readers 7–9 viewed images processed by contrast-limited adaptive histogram equalization (CLAHE).[26] CLAHE used a clip limit of 2.0 and an 8×8 tile

grid on the lightness channel. Cross-provider comparisons combined existing self-consistency-reader responses.

Reference cases were retrieved in-context (few-shot) examples: closed question–answer pairs from training images. They were evaluated on VQA-RAD and SLAKE using VQA-RAD and English SLAKE training images with closed yes/no questions. The data sets carried no patient identifiers; reference cases were excluded by image identity and similarity. Images identical to any test image were excluded by Secure Hash Algorithm 256 (SHA-256) hash. For each query, exclusions also removed reference cases within Hamming distance 8 of its 64-bit perceptual hash or with CLIP cosine similarity of at least 0.98.

Retrieval used CLIP ViT-L/14-336 image embeddings. The 27 nearest eligible images were shuffled deterministically by question identifier into 9 groups of 3. For shared reference cases, the first group was supplied to every reader; for private reference cases, group i was supplied to reader i. Reference images accompanied question–answer pairs selected by maximal term frequency–inverse document frequency (TF-IDF) cosine similarity to the current question. Text-only reference cases supplied the same question and answer without the reference image. Single readers also received reference cases with images or as text.

2.8. Experiment D: open-weight models

Open-weight models received the same rendered pixels as the proprietary models. Conditions included original and blank images, 256- and 128-pixel images, gamma 0.7, CLAHE, and target and control occlusion on SLAKE-target. SLAKE-pairs used original images only. Each item and condition required one forward pass without sampling. Option probabilities were obtained by applying softmax to next-token logits for the answer words yes/no on binary items and option letters otherwise. The highest logit among accepted capitalization and leading-space token variants represented each answer.

2.9. Outcome measures

Accuracy was the proportion of answers matching the reference. Error correlation, ϕ, was the mean Pearson correlation of binary error indicators across items over reader pairs. Pairs with constant error indicators were omitted. N_{eff} was calculated as $N_{\text{eff}} = 9/(1 + 8\phi)$.[27] For readers of unequal

Table 2: Experiments, conditions, readers, and models. Proprietary models were GPT-5 mini, Gemini 3.7 Flash, Claude Haiku 4.5, and Qwen 3.8 Flash. Qwen 3.8 Flash ran A0, A1, the self-consistency panel, and contrast adjustment; the other proprietary conditions used the first three models. CLAHE indicates contrast-limited adaptive histogram equalization.

Experiment	Data sets	Conditions	Readers	Models
A0, original images	VQA-RAD, SLAKE, MedXpertQA-MM	Original image	Single reader; independent panels of 1, 3, 9; discussion panels of 3, 9	4 proprietary
A1, reduced resolution and blank images	VQA-RAD, SLAKE, MedXpertQA-MM	256 and 128 pixels; blank image	Single reader; independent panels	4 proprietary
A2, region occlusion	SLAKE-target	Target box; control box	Single reader; independent panels	3 proprietary
A3, paired images	SLAKE-pairs	Original images	Single reader; independent panels	3 proprietary
B, perception and interpretation	VQA-RAD, SLAKE, MedXpertQA-MM	Query-agnostic findings reports; own-report or shared-report condition	Panels of 9 readers answering from reports	3 perception $\times$ 3 interpretation
C, shared components	VQA-RAD, SLAKE, MedXpertQA-MM	Self-consistency panel; contrast adjustment (original, gamma 0.7, CLAHE); reference cases on VQA-RAD and SLAKE (shared or private, with images or as text)	Self-consistency panels of 9; single reader with reference cases	4 proprietary; reference cases: 3
D, open-weight models	VQA-RAD, SLAKE, MedXpertQA-MM, SLAKE-target, SLAKE-pairs	Original, blank, reduced resolution, contrast adjustment; occlusion on SLAKE-target; original images only on SLAKE-pairs	One forward pass per item	5 open-weight

accuracy, correlation was also divided by its maximum given their marginal error rates, $\phi_{\max}$.

The gain over the single reader was divided into two parts that summed to it. The first was the mean accuracy of the panel's readers minus the single reader's accuracy. The second was majority-vote accuracy minus the mean accuracy of the panel's readers (aggregation). For all panels of 9 readers on yes/no items, observed majority-vote accuracy was compared with accuracy predicted by a beta-binomial model. The model's mean was the readers' mean accuracy, and its intraclass correlation was their error correlation. A majority of 9 required at least 5 correct readers. Illustrative beta-binomial curves used mean reader accuracies of 60%, 75%, and 90%.

Each reader's correctness with original and blank images was cross-tabulated. Shared wrong answers were items that at least 7 of 9 readers answered wrongly; their majority answers were compared with blank-image majorities.

Answer-change rate was the proportion of answers changed by occlusion. Occlusion sensitivity was the target-occlusion answer-change rate minus the control-occlusion answer-change rate, calculated for readers and panel majorities.[28] Pair accuracy required both paired answers to be correct; the proportion of pairs answered differently was also measured. Paired answers were compared as option text (yes or no).

The report conditions (Experiment B) were compared by accuracy and error correlation, and errors were separated by report coverage of the queried target. Reader and panel accuracies from reports were compared with those of the self-consistency panel of the same interpretation model reading images directly.

Matched contrasts of mean pairwise error correlation were the primary analysis of Experiment C. The seven contrasts were defined as follows:

- Role: independent panel minus self-consistency panel.
- Contrast adjustment: readers viewing different contrast adjustments minus self-consistency readers viewing original images.
- Reference cases: private minus shared reference cases.
- Report: own-report minus shared-report condition, using same-model reports.

- Perception model: reports from different perception models minus different reports from one perception model, within an interpretation model.
- Interpretation model: two interpretation models reading the same report minus one interpretation model reading one shared report.
- Model: self-consistency readers from two models minus self-consistency readers from one model.

Contrasts were computed within each interpretation model or model pair and then averaged over models. For the interpretation-model and model contrasts, the subtracted mean was averaged over the two models. The reference error correlation came from self-consistency readers of one model.

An ordinary least squares regression was an additive secondary summary of pairwise error correlation. Predictors indicated the components two readers did not share; the regression included one intercept per pair of interpretation models.

Discussion analyses compared initial and final error correlation. Conformity was the proportion of readers correct in their initial answer whose final answer was wrong.[29,30] It was stratified by whether the initial answer differed from the same reader's blank-image answer and by whether it was in the initial majority.

For open-weight models, visual reliance was the absolute change in log-odds of yes between original and blank images. For multiple-option items, visual reliance was the Kullback–Leibler divergence from the original-image to blank-image probability distribution. Model-pair error correlations were compared by shared vision encoder, and Spearman correlation related proprietary-model shared wrong answers to open-weight visual reliance.

2.10. Statistical analysis

Percentile bootstrap 95% confidence intervals (CIs) used 2,000 item-level resamples with seed 0. Error-correlation intervals used 500 resamples; the regression, matched contrasts, and comparison of initial with final error correlation used 300 resamples. Matched contrasts used the same resampled items as the regression. All differences were paired by item; resampling retained all reader responses and matched conditions for each item. Paired-image measures resampled complete pairs.

Abstentions were answers that could not be parsed, mostly refusals to read the blank image. They counted as wrong and were reported for each condition. A sensitivity analysis computed error correlation on items both readers answered. Results are reported as estimates with 95% CIs. Differences whose 95% CI excluded zero are described as increases or decreases.

2.11. Validation before the main experiments

Six validation checks fixed the settings of the main experiments (Supplementary Table S1).

2.12. Code and data availability

The code is public at `https://github.com/hq0709/consultation-arithmetic`. VQA-RAD, SLAKE, and MedXpertQA are public data sets.

3. Results

3.1. Reading panels and error correlation

For GPT-5 mini, Gemini 3.7 Flash, and Claude Haiku 4.5, independent panels changed accuracy by −2.8 to 4.0 percentage points relative to the single reader. Model-specific ranges were −2.8 to 0.8, −1.2 to 1.2, and −2.8 to 4.0 percentage points, respectively (Fig. 2a). Their discussion-panel differences ranged from −3.2 to 5.6 percentage points. The Claude Haiku 4.5 discussion panel increased accuracy on VQA-RAD by 5.6 percentage points (95% CI, 0.8 to 10.8).

Qwen 3.8 Flash independent panels increased accuracy on VQA-RAD by 13.6 percentage points (95% CI, 7.6 to 19.6). Gains were 8.0 percentage points (95% CI, 4.0 to 12.4) on SLAKE and 5.6 percentage points (95% CI, 0.4 to 11.2) on MedXpertQA-MM. Discussion-panel gains ranged from 8.8 to 14.0 percentage points (Fig. 2a; Supplementary Tables S2–S5).

For Qwen 3.8 Flash, mean reader accuracy exceeded single-reader accuracy by 3.1 to 10.3 percentage points; majority voting added 1.1 to 3.3 percentage points. For the other models, majority voting changed accuracy by −2.1 to 2.4 percentage points relative to mean reader accuracy (Supplementary Tables S2–S5).

Panels of nine readers had an N_{eff} of 1.17 to 1.99 (Fig. 2b). Values ranged from 1.17 to 1.39 for GPT-5 mini, Gemini 3.7 Flash, and Claude Haiku 4.5, and from 1.46 to 1.99 for Qwen 3.8 Flash.

After discussion, error correlations were 0.995 to 1.000 for GPT-5 mini, Gemini 3.7 Flash, and Claude Haiku 4.5 (Fig. 2c). Qwen 3.8 Flash correlations were 0.822 to 0.913. Unanimity after discussion ranged from 82.4% to 100.0%, and 8.9% to 70.0% of unanimous answers were wrong.

Across 108 panels on yes/no items, observed majority accuracy minus the beta-binomial prediction averaged −0.07 percentage points (Fig. 2d). The standard deviation (SD) was 0.84 percentage points, and the largest absolute difference was 2.59 percentage points. Majority accuracy minus mean reader accuracy ranged from −2.1 to 5.6 percentage points across error correlations of 0.44 to 0.99 (Fig. 2e).

3.2. Visual evidence and the language prior

Blank images lowered single-reader accuracy by 11.2 to 26.8 percentage points on VQA-RAD and 25.2 to 40.0 percentage points on SLAKE (Fig. 3a). Accuracy ranged from 60.8% to 81.2% with original VQA-RAD images and from 46.8% to 60.8% with blank images. Corresponding SLAKE ranges were 72.4% to 91.2% and 38.8% to 54.4%. Independent-panel accuracy also decreased on both data sets (Supplementary Tables S6–S8).

On MedXpertQA-MM, original-image minus blank-image single-reader accuracy was 4.0 percentage points (95% CI, −2.0 to 10.0) for GPT-5 mini. It was 3.6 percentage points (95% CI, −0.4 to 8.0) for Claude Haiku 4.5. Qwen 3.8 Flash had a difference of 2.0 percentage points (95% CI, −4.4 to 8.0). With blank images, Gemini 3.7 Flash accuracy decreased by 18.0 percentage points (95% CI, 12.0 to 23.6). Independent-panel accuracy decreased for GPT-5 mini, Gemini 3.7 Flash, and Qwen 3.8 Flash (Supplementary Tables S6–S8).

For Gemini 3.7 Flash, Claude Haiku 4.5, and Qwen 3.8 Flash, blank-image error correlations were 0.841 to 0.922 on VQA-RAD and 0.831 to 0.982 on SLAKE (Fig. 3b). Corresponding original-image ranges were 0.439 to 0.731 and 0.529 to 0.785. On VQA-RAD, GPT-5 mini error correlation fell from 0.752 with original images to 0.694 at 256 pixels and 0.644 at 128 pixels. Qwen 3.8 Flash correlations rose from 0.439 to 0.442 and 0.480, respectively.

After abstentions were excluded, Claude Haiku 4.5 blank-image error correlation was 0.963 (95% CI, 0.942 to 0.982) on VQA-RAD. It was 0.931 (95% CI, 0.901 to 0.956) on SLAKE and 0.852 (95% CI, 0.803 to 0.891) on MedXpertQA-MM. Abstentions ranged from 4.3% to 14.8% across data sets.

On VQA-RAD, 24.1% to 37.3% of reader answers were correct only with the original image, compared with 10.4% to 13.6% only with the blank image

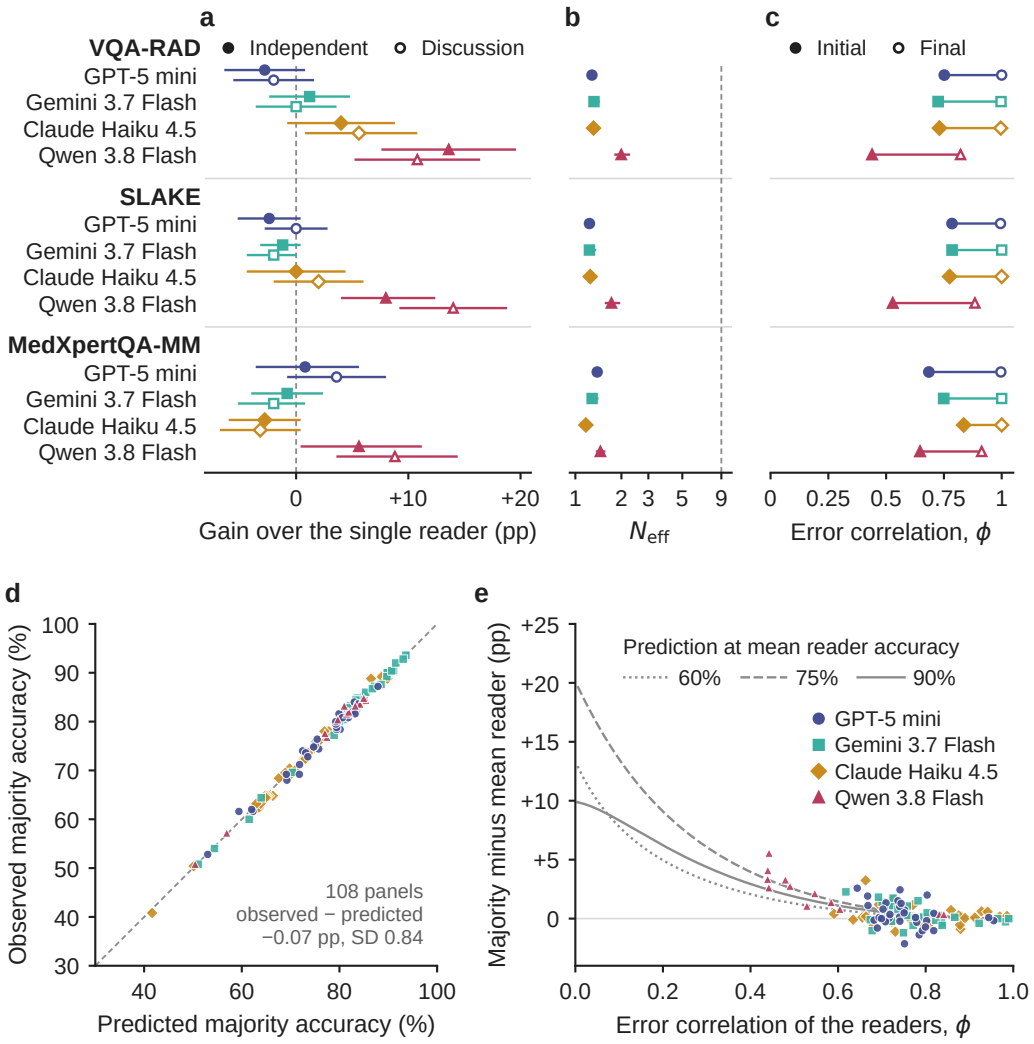


Figure 2: Reading panels and error correlation. (a) Gain over the single reader (pp): Independent (filled) and Discussion (open), each with 9 readers. (b) Effective number of independent readers, N_{eff}. (c) Error correlation, ϕ: Initial (filled) and Final (open). Rows group VQA-RAD, SLAKE, and MedXpertQA-MM (250 items each). (d) Observed versus beta-binomial predicted majority accuracy across 108 panels on VQA-RAD, SLAKE, SLAKE-target (250 items each), and SLAKE-pairs (200 pairs). Observed minus predicted accuracy was −0.07 percentage points, SD 0.84. (e) Majority minus mean reader accuracy; Prediction at mean reader accuracy identifies 60% (dotted), 75% (dashed), and 90% (solid) curves; the model legend is inside the panel. Blue circles denote GPT-5 mini; teal squares, Gemini 3.7 Flash. Amber diamonds denote Claude Haiku 4.5; rose triangles, Qwen 3.8 Flash. Item bootstrap 95% CIs are paired (a) or transformed (b). pp indicates percentage points.

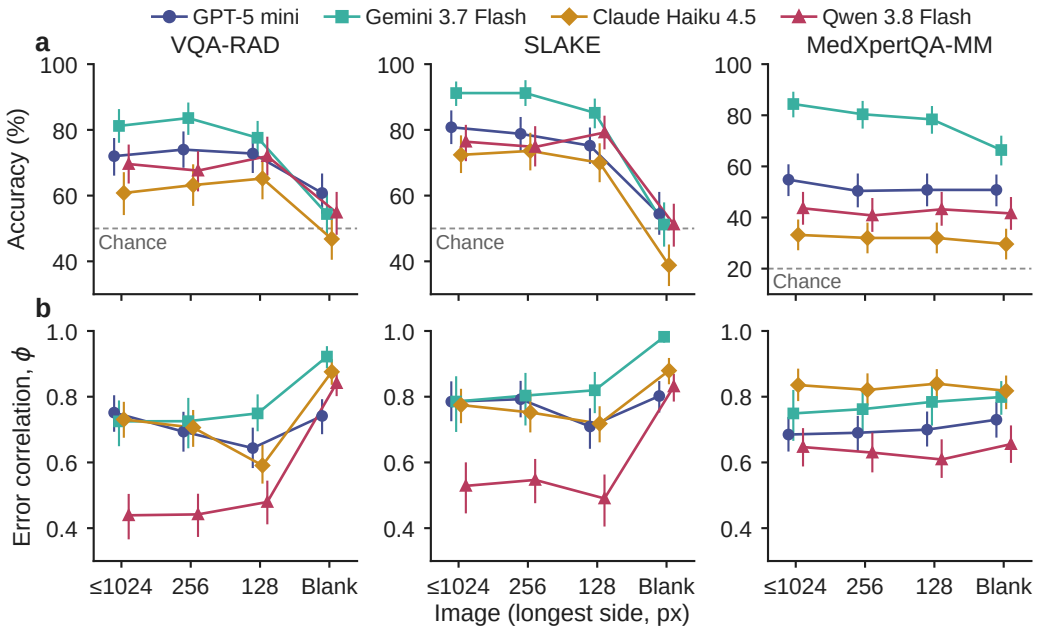


Figure 3: Accuracy and error correlation with reduced resolution and blank images. (a) Single-reader accuracy. (b) Error correlation, ϕ, among independent-panel readers. Columns show VQA-RAD, SLAKE, and MedXpertQA-MM (250 items each). Image conditions are labeled $\leq$ 1024, 256, 128, and Blank. Pixel caps apply to the longest side; blank images are uniform gray images of unchanged dimensions. Blue circles denote GPT-5 mini; teal squares, Gemini 3.7 Flash. Amber diamonds denote Claude Haiku 4.5; rose triangles, Qwen 3.8 Flash. Lines join adjacent image conditions. Dashed horizontal lines in (a) mark chance accuracy: 50.0% for VQA-RAD and SLAKE and 20.0% for MedXpertQA-MM. Vertical intervals are item bootstrap 95% CIs. px indicates pixels.

(Fig. 4a). Corresponding SLAKE ranges were 36.3% to 46.4% and 4.4% to 15.8%. On MedXpertQA-MM, they were 8.1% to 22.3% and 4.4% to 7.9%; 12.3% to 64.0% of answers were wrong with both images.

On MedXpertQA-MM, 64.2% to 76.2% of shared wrong answers matched the blank-image majority answer (Fig. 4b). The corresponding shares were 42.9% to 54.4% on VQA-RAD and 14.3% to 57.9% on SLAKE.

3.3. Occlusion sensitivity and paired images

On SLAKE-target, occlusion sensitivity was 17.1 percentage points (95% CI, 13.2 to 21.2) for GPT-5 mini (Fig. 5a,b). It was 32.9 percentage points (95% CI, 28.2 to 38.0) for Gemini 3.7 Flash. Claude Haiku 4.5 occlusion

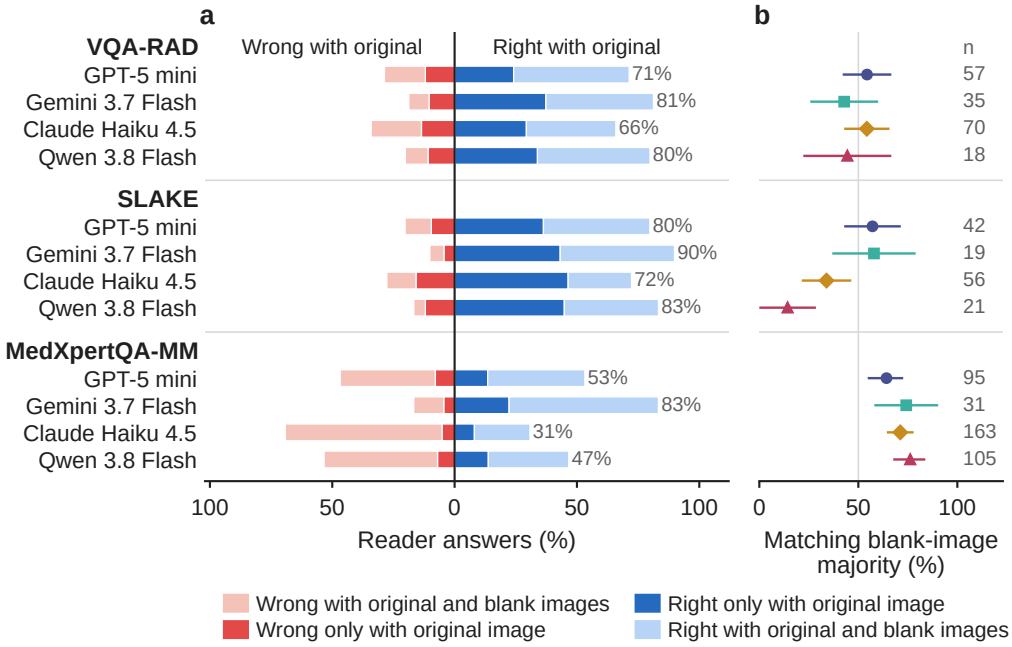


Figure 4: Reader correctness with original and blank images and shared wrong answers matching the blank-image majority. (a) Reader answers from independent panels of 9, cross-tabulated by correctness with each image. The headers are Wrong with original and Right with original. (b) Matching blank-image majority (%) is the proportion of shared wrong answers matching the blank-image majority. Shared wrong answers are items that at least 7 of 9 readers answered wrongly. Data sets are VQA-RAD, SLAKE, and MedXpertQA-MM (250 items each). In (a), light red denotes Wrong with original and blank images; dark red, Wrong only with original image. Dark blue denotes Right only with original image; light blue, Right with original and blank images. End labels give original-image accuracy. In (b), blue circles denote GPT-5 mini; teal squares, Gemini 3.7 Flash. Amber diamonds denote Claude Haiku 4.5; rose triangles, Qwen 3.8 Flash. The vertical zero line in (a) separates wrong answers with the original image on the left from right answers on the right. The vertical reference in (b) marks 50.0%. Horizontal intervals in (b) are item bootstrap 95% CIs. n indicates the number of shared wrong answers.

sensitivity was 3.2 percentage points (95% CI, 0.8 to 5.7).

Open-weight occlusion sensitivity ranged from 2.4 to 21.6 percentage points (Fig. 5b; Supplementary Tables S11–S12). The LLaVA-1.5 7B estimate was 2.4 percentage points (95% CI, −2.0 to 6.8). Target occlusion increased answer changes relative to control occlusion for the other four models.

Gemini 3.7 Flash had the highest paired-image accuracy among proprietary models: 74.0% (95% CI, 67.5 to 80.0) on SLAKE-pairs (Fig. 5c). Both single-reader answers were correct for 59.0% (95% CI, 52.0 to 65.5) of pairs with GPT-5 mini. The rate was 38.5% (95% CI, 31.5 to 45.0) with Claude Haiku 4.5. Answers differed between paired images in 62.0%, 75.0%, and 42.5% of pairs for GPT-5 mini, Gemini 3.7 Flash, and Claude Haiku 4.5, respectively (Supplementary Tables S13–S14).

3.4. Perception and interpretation

In the own-report condition, Gemini 3.7 Flash reports yielded 78.7% to 81.3% accuracy on VQA-RAD and 84.3% to 88.2% on SLAKE across interpretation models (Fig. 6a). Corresponding ranges with GPT-5 mini reports were 69.0% to 70.6% and 81.0% to 84.9%. With Claude Haiku 4.5 reports, they were 61.6% to 63.6% and 76.3% to 80.2%.

With Gemini 3.7 Flash reports, accuracy gains ranged from 6.1 percentage points (95% CI, 1.4 to 10.9) to 16.3 percentage points (95% CI, 9.9 to 22.6). These gains were relative to the other interpretation models reading the images themselves. Claude Haiku 4.5 reports lowered the other interpretation models' accuracy in 5 of 6 cells, by 7.9 to 25.9 percentage points. The sixth difference, for GPT-5 mini on SLAKE, was −0.4 percentage points (95% CI, −5.0 to 4.6). Accuracy with same-model reports increased only for GPT-5 mini on SLAKE: 4.4 percentage points (95% CI, 0.9 to 8.2). Gemini 3.7 Flash had a paired difference of −9.2 percentage points (95% CI, −13.3 to −5.2) with same-model reports on MedXpertQA-MM.

On MedXpertQA-MM, Gemini 3.7 Flash achieved 60.8%, 74.5%, and 57.8% accuracy with GPT-5 mini, Gemini 3.7 Flash, and Claude Haiku 4.5 reports, respectively. Claude Haiku 4.5 achieved 30.6%, 45.6%, and 27.4% with the same reports; GPT-5 mini achieved 50.1%, 62.0%, and 45.0%. Image-reading accuracy was 83.7%, 31.7%, and 52.9% for Gemini 3.7 Flash, Claude Haiku 4.5, and GPT-5 mini, respectively.

With same-model reports, error correlations were 0.800 to 0.990 in the shared-report condition, compared with 0.641 to 0.818 in the own-report

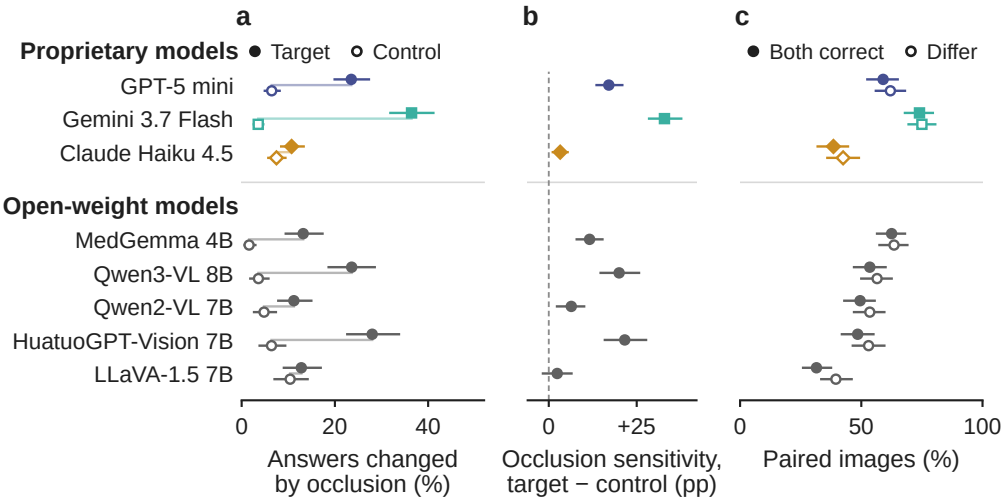


Figure 5: Occlusion sensitivity and paired images, grouped as Proprietary models and Open-weight models. (a) Answers changed by occlusion (%) on SLAKE-target (250 items), labeled Target and Control. (b) Occlusion sensitivity, target − control (pp), is the difference between the two answer-change rates. (c) Paired images (%) on SLAKE-pairs (200 pairs with identical questions and opposite reference answers), labeled Both correct and Differ. Proprietary-model occlusion estimates average panel-reader responses within items; paired-image estimates use single readers. Open-weight models use one forward pass per item. Filled and open markers denote Target and Control in (a), and Both correct and Differ in (c). Blue circles denote GPT-5 mini; teal squares, Gemini 3.7 Flash. Amber diamonds denote Claude Haiku 4.5; dark gray circles denote the labeled open-weight models. The dashed vertical line in (b) marks zero. Horizontal intervals are item bootstrap 95% CIs, with complete pairs resampled in (c). Occlusion sensitivity is expressed in percentage points.

condition (Fig. 6b). Image-reading correlations ranged from 0.692 to 0.915. Shared-report correlation exceeded image-reading correlation except for Claude Haiku 4.5 on MedXpertQA-MM (0.893 versus 0.915).

The queried SLAKE target was omitted in 25.8% to 47.1% of reports behind an error, compared with 26.5% to 30.8% of all reports. With Gemini 3.7 Flash as the perception model, omission was more frequent among reports behind errors (33.6% to 47.1%) than among all reports (26.7%).

3.5. Error correlation by shared component

For self-consistency readers, error correlation ranged from 0.442 to 0.629 for Qwen 3.8 Flash and from 0.692 to 0.800 for GPT-5 mini (Fig. 7a). Ranges

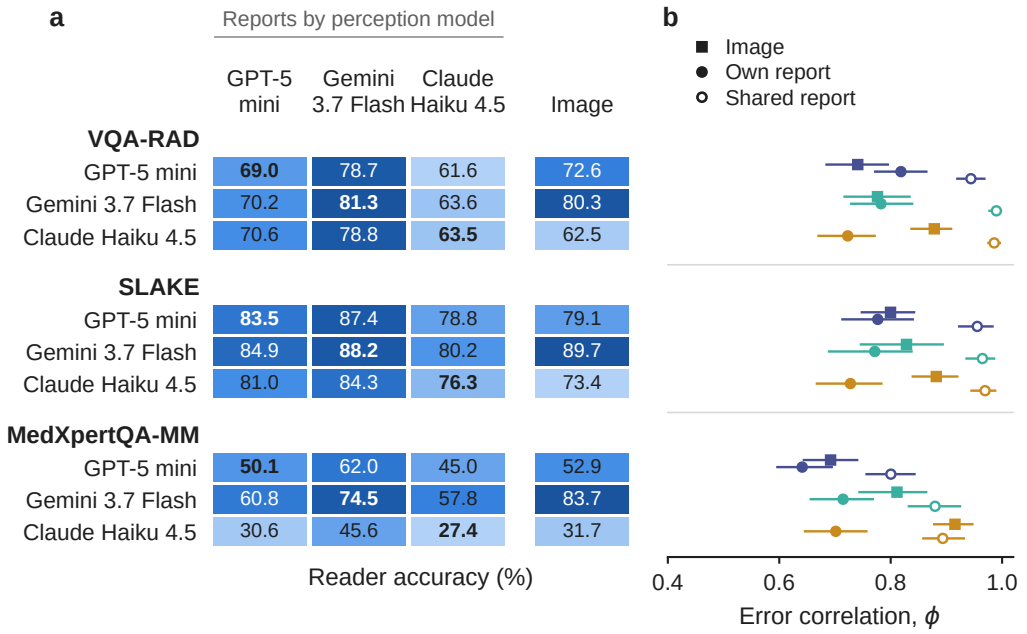


Figure 6: Reader accuracy and error correlation with query-agnostic findings reports, written without the question or answer options. (a) Mean reader accuracy in the own-report condition, where each reader read a different report without the image, or when reading the image directly (Image column). Under Reports by perception model, columns identify perception models; rows identify interpretation models within VQA-RAD, SLAKE, and MedXpertQA-MM (250 items each). Cell numbers give accuracy percentages; darker blue denotes higher accuracy within each data set. Bold numbers identify matching perception and interpretation models. (b) Error correlation, ϕ, among 9 readers. Filled squares denote Image; filled circles, Own report with same-model reports; open circles, Shared report with one report from that model. Blue denotes GPT-5 mini; teal, Gemini 3.7 Flash; amber, Claude Haiku 4.5. Horizontal intervals are item bootstrap 95% CIs. Light gray horizontal lines separate data sets.

were 0.776 to 0.828 for Gemini 3.7 Flash and 0.878 to 0.915 for Claude Haiku 4.5.

Changing the model lowered mean error correlation by 0.470 to 0.492 across data sets (Fig. 7b; Supplementary Tables S20–S22 give estimates with 95% CIs). Perception-model contrasts ranged from −0.311 to −0.263, and interpretation-model contrasts from −0.430 to −0.108. Within-model contrasts ranged from −0.060 to −0.033 for role and from −0.094 to −0.045 for contrast adjustment. Ranges were −0.146 to −0.110 for reference cases and −0.204 to −0.172 for report.

On VQA-RAD, per-model contrasts ranged from −0.148 to 0.011 for role and from −0.181 to 0.023 for contrast adjustment. Reference-case contrasts ranged from −0.229 to −0.043 on VQA-RAD and from −0.247 to −0.082 on SLAKE. MedXpertQA-MM interpretation-model contrasts ranged from −0.546 to −0.310; VQA-RAD model contrasts ranged from −0.696 to −0.272.

In the regression, an additive secondary summary, the VQA-RAD report coefficient was 0.150 (95% CI, 0.092 to 0.218), compared with a matched contrast of −0.199. Different perception and interpretation models were associated with lower error correlation across data sets (Supplementary Tables S23–S24).

3.6. Reference cases

On VQA-RAD, Claude Haiku 4.5 accuracy was 60.8% without reference cases, 68.8% with reference images, and 61.2% with reference text (Fig. 8a). On SLAKE, its corresponding accuracies were 72.4%, 67.6%, and 76.8%. GPT-5 mini and Gemini 3.7 Flash achieved 73.6% and 82.4% with reference images on VQA-RAD, compared with 75.6% and 83.6% with reference text. Corresponding SLAKE accuracies were 79.2% and 89.2% with images, and 81.6% and 90.8% with text.

Private reference cases lowered error correlation in every model and format, from 0.741 to 0.667 for GPT-5 mini on SLAKE with reference images (Fig. 8c). Its panel accuracy was 81.6% with shared reference images and 84.0% with private reference images (Fig. 8b). Gemini 3.7 Flash panel accuracy was 90.4% with shared reference text and 92.0% with private reference text; error correlation was 0.823 and 0.716.

For Claude Haiku 4.5, private reference cases yielded equal or lower panel accuracy in both formats and data sets (Fig. 8b). With SLAKE reference images, error correlation was 0.928 with shared cases and 0.655 with private cases; panel accuracy was 70.4% and 68.8%, respectively. With reference text, error correlation was 0.941 and 0.720, while panel accuracy remained 78.0%.

3.7. Discussion rounds

On MedXpertQA-MM, conformity ranged from 7.2% to 23.6% when correct initial answers differed from the blank-image answer, compared with 2.6% to 6.3% when they matched it (Fig. 9a). Paired differences ranged

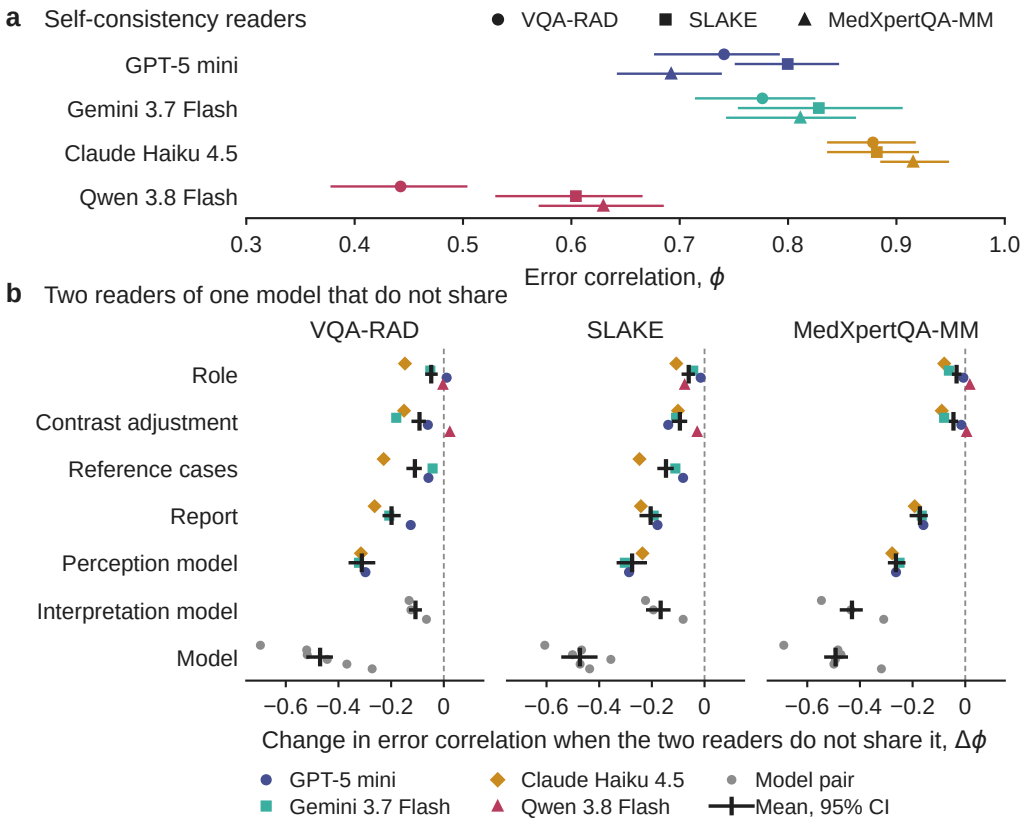


Figure 7: Error correlation by shared reader component. (a) Self-consistency readers: error correlation, ϕ, between two readers of one model. Circles denote VQA-RAD; squares, SLAKE; triangles, MedXpertQA-MM (250 items each). (b) Matched differences in mean pairwise error correlation, $\Delta\phi$, for pairs differing in the named component relative to pairs sharing it. Columns identify data sets. Role compares independent with self-consistency panels. Contrast adjustment compares different adjustments with original images. The Reference cases row compares private with shared cases on VQA-RAD and SLAKE; Report compares the own-report with the shared-report condition. Perception model compares reports from different perception models with different reports from one perception model. Interpretation model compares different interpretation models with one interpretation model; all read a shared report. Model compares different models with one model; all read images. Blue circles denote GPT-5 mini; teal squares, Gemini 3.7 Flash. Amber diamonds denote Claude Haiku 4.5; rose triangles, Qwen 3.8 Flash. Gray circles denote model pairs; black ticks, means. Horizontal intervals in (a) and black intervals in (b) are item bootstrap 95% CIs. Dashed lines mark zero change.

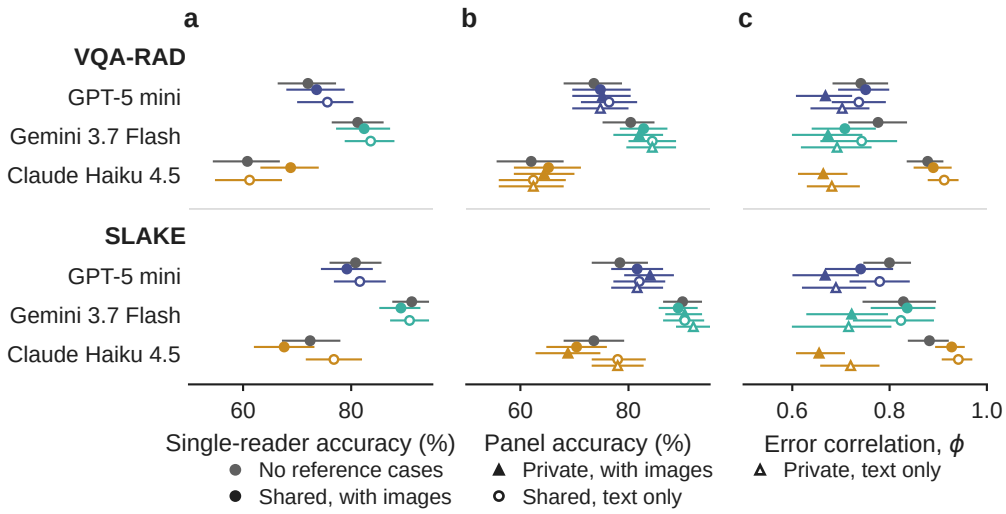


Figure 8: Accuracy and error correlation with reference cases. (a) Single-reader accuracy with no reference cases, with reference images and question–answer pairs, or with the same pairs as text only. (b) Majority-vote accuracy of self-consistency panels of 9 readers. (c) Panel error correlation, ϕ. Rows group VQA-RAD and SLAKE (250 items each) by model. Gray filled circles denote No reference cases. Filled colored circles denote Shared, with images; filled triangles, Private, with images. Open colored circles denote Shared, text only; open triangles, Private, text only. Private reference cases were assigned separately to each reader. Blue denotes GPT-5 mini; teal, Gemini 3.7 Flash; amber, Claude Haiku 4.5. Horizontal intervals are item bootstrap 95% CIs. Light gray horizontal lines separate data sets.

from 4.6 percentage points (95% CI, 0.8 to 9.0) for Gemini 3.7 Flash to 17.4 percentage points (95% CI, 6.4 to 29.7) for Claude Haiku 4.5.

On VQA-RAD, conformity was 5.1 percentage points higher (95% CI, 0.4 to 10.4) for GPT-5 mini when initial answers differed from the blank-image answer (Fig. 9a). The difference was 5.9 percentage points (95% CI, 0.6 to 11.5) for Qwen 3.8 Flash. On SLAKE, conformity was 2.6 percentage points higher (95% CI, 0.3 to 5.2) for GPT-5 mini.

Readers changed 41.5% to 100.0% of their correct initial minority answers to a wrong answer, compared with 1.0% to 6.5% of correct initial majority answers (Fig. 9b).

On MedXpertQA-MM, 6.6% to 23.5% of correct initial answers that differed from the blank-image answer were in the initial minority (Fig. 9c). The corresponding shares were 2.5% to 11.2% on VQA-RAD and 1.8% to 4.0%

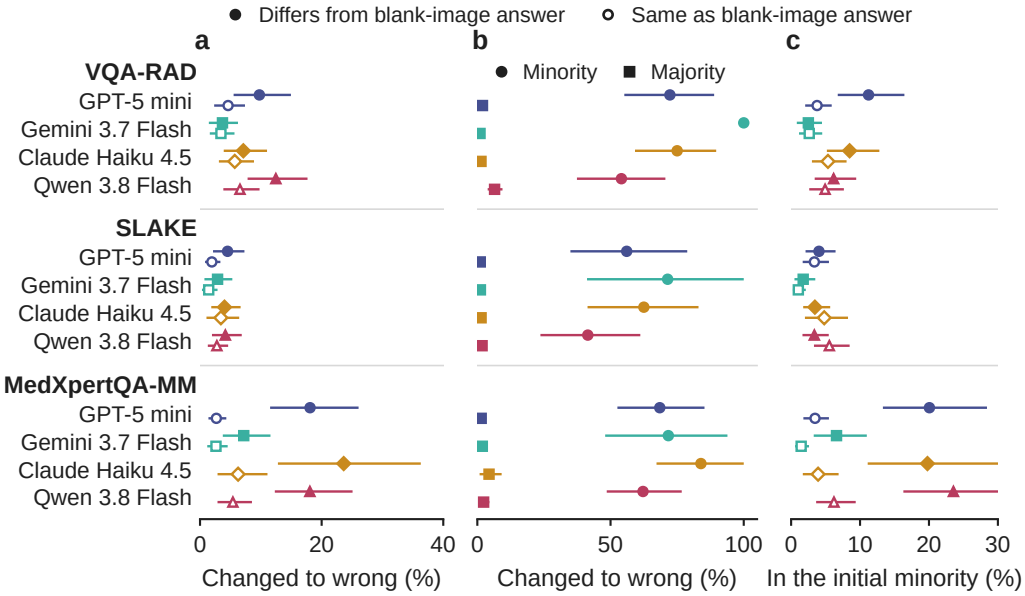


Figure 9: Correct initial answers changed to wrong answers during discussion. (a) Conformity, the proportion of readers with correct initial answers whose final answers were wrong, stratified by the same reader's blank-image answer. (b) Conformity among correct initial minority and majority answers. Both axes are labeled Changed to wrong (%). (c) Proportion of correct initial answers in the initial minority, labeled In the initial minority (%). Discussion panels contain 9 readers. Rows group VQA-RAD, SLAKE, and MedXpertQA-MM (250 items each). Filled and open markers in (a) and (c) denote Differs from blank-image answer and Same as blank-image answer, respectively. Circles and squares in (b) denote Minority and Majority, respectively, for the initial answers. Blue denotes GPT-5 mini; teal, Gemini 3.7 Flash. Amber denotes Claude Haiku 4.5; rose, Qwen 3.8 Flash. In (a) and (c), model markers are circles, squares, diamonds, and triangles, respectively. Estimates are ratios of reader counts; horizontal intervals are item bootstrap 95% CIs.

on SLAKE.

3.8. Open-weight models

With blank images, four of five open-weight models lost 16.4 to 19.2 percentage points on VQA-RAD; LLaVA-1.5 7B lost 2.0 percentage points. All five lost 16.0 to 48.8 percentage points on SLAKE. Original-image accuracy ranged from 56.0% to 80.0% on VQA-RAD and from 66.0% to 88.8% on SLAKE (Supplementary Fig. S1a). Blank-image ranges were 54.0% to 60.8% and 40.0% to 50.8%, respectively. On MedXpertQA-MM, original-image ac-

curacy ranged from 20.0% to 24.4%, compared with 20.0% to 23.6% with blank images. MedGemma 4B had the highest original-image accuracy on VQA-RAD and SLAKE; Qwen3-VL 8B had the highest on MedXpertQA-MM.

Mean absolute change in log-odds of yes ranged from 0.89 to 6.43 on VQA-RAD and from 1.40 to 11.23 on SLAKE (Supplementary Fig. S1b). On MedXpertQA-MM, mean visual reliance measured by Kullback–Leibler divergence ranged from 0.10 to 0.88.

HuatuoGPT-Vision 7B and LLaVA-1.5 7B, the pair sharing a vision encoder, had error correlation divided by its maximum given marginal error rates of 0.213 to 0.451 (Supplementary Fig. S1c). Values for pairs with different encoders ranged from 0.148 to 0.725.

The Spearman correlation between shared wrong answers of proprietary models and open-weight visual reliance was −0.401 (95% CI, −0.502 to −0.295) on SLAKE. It was −0.048 (95% CI, −0.165 to 0.068) on VQA-RAD and 0.006 (95% CI, −0.128 to 0.136) on MedXpertQA-MM.

4. Discussion

Correlated errors limited the contribution of additional readers built from one VLM (Fig. 2b). These measurements extend the finding of correlated errors across language models to reading panels answering questions about radiology images.[9] In a companion analysis of text-only clinical questions, error correlation was of the same magnitude as the imaging values and rose during discussion (unpublished data; Supplementary Section S5).

The beta-binomial fit explains the observed majority-vote accuracy on yes/no items in terms of mean reader accuracy and error correlation (Fig. 2d,e). High error correlation limits the contribution of majority voting because readers tend to be wrong on the same items. Qwen 3.8 Flash's larger gain over the single reader came mainly from more accurate panel readers; the aggregation gain was smaller (Fig. 2a,e).

Conformity during discussion reduced the contribution of correct minority answers to the final consensus (Fig. 9). Readers changed correct initial answers to a wrong answer when they revised their responses after seeing peer answers. This pattern is consistent with conformity in multi-agent debate.[30] Automation bias in mammography is a clinical parallel: radiologists accepted incorrect AI suggestions for Breast Imaging Reporting and Data System categories.[31]

The language prior contributed to correlated errors on MedXpertQA-MM, where shared wrong answers often matched blank-image majorities (Fig. 4). Agreement with the blank-image answer identifies decisions that persisted when visual findings were unavailable. The influence of question text parallels the effect of clinical information on diagnostic test interpretation in Loy and Irwig's systematic review.[32]

Occlusion sensitivity identified differences among models in their dependence on the queried anatomy (Fig. 5a,b). The comparison with an equal-sized control region distinguishes sensitivity to the queried anatomy from sensitivity to occlusion elsewhere.[28] Gemini 3.7 Flash's stronger occlusion sensitivity indicates greater dependence on the queried anatomy than in the other proprietary models. Single readers' answers differed between paired images in 42.5% to 75.0% of pairs, although every pair had opposite reference answers (Fig. 5c). Blank-image accuracy losses and occlusion sensitivity in the open-weight probes show that these effects also occurred without sampling (Fig. 5b; Supplementary Fig. S1a).

Perception and interpretation each contributed to accuracy in the report conditions (Fig. 6a). Separating these stages, as in Prism, allowed the same findings to be evaluated by different interpretation models.[18] Gemini 3.7 Flash reports improved the other interpretation models' accuracy, while accuracy differed among models reading identical reports. Higher error correlation in the shared-report condition than in the own-report condition identifies common findings as a source of correlated errors (Fig. 6b). A query-agnostic findings report makes the extracted findings available for inspection; its omissions restrict the visual information available to the interpretation model.

Changing the model reduced error correlation most, which supports using a different model for a second reading (Fig. 7b). Changing the perception model ranked next on VQA-RAD and SLAKE. On MedXpertQA-MM, changing the interpretation model reduced error correlation more than changing the perception model. On VQA-RAD and SLAKE, within-model reductions were ordered from smallest to largest as roles, contrast adjustments, reference cases, and reports. The comparison of shared and private cases supports assessing a second reading by the errors it corrects and introduces (Fig. 8). For Claude Haiku 4.5, lower error correlation accompanied equal or lower panel accuracy.

This evaluation concerned closed questions from public data sets. Its scope did not include human readers or clinical reporting.

5. Conclusions

For three proprietary VLMs, independent and discussion panels changed accuracy little relative to the single reader. Qwen 3.8 Flash gained more, mainly through more accurate panel readers. Error correlation limited nine-reader panels to an N_{eff} of 1.17 to 1.99. Language priors contributed to correlated errors on MedXpertQA-MM; the shared-report condition increased error correlation relative to the own-report condition. Conformity during discussion was greater for correct initial minority answers than for correct initial majority answers. A second AI reading lowered error correlation most when it came from a different model.

Authors' contribution

TBD.

Ethics statement

The study used only public, de-identified data sets and involved no human participants. Institutional review board approval was not required.

Conflicts of interest

The authors declare no conflicts of interest.

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